

THI THANH HUONG VO

Email: vohuong2702@gmail.com

<https://orcid.org/0009-0002-2561-954X> / <https://www.linkedin.com/in/thanhhuongvo/>

Address: Ho Chi Minh city, Vietnam / Gyeonggi, Republic of Korea

Objective: Seeking opportunities to pursue a PhD degree in Materials Sciences Engineering

Keywords: Wide bandgap semiconductor, Reliability, Experiment, Electrical and Optoelectronic

English

Listening



Speaking



Reading



Writing



Data Analysis

Origin



MATLAB



Microsoft



LATEX



Simulation

TCAD



AutoCAD



MATLAB



Experiment

Electrical



Photonics



Spectroscopy



About me

Independent
Thorough
Energetic
Flexible

Education background

Korea Aerospace University – Master of Science – September 2022 to February 2025

- Department of Materials Science and Engineering
- Department of Smart Air Mobility
- Major: Drone Materials and Components
- GPA: 4.0/4.5
- Graduation thesis: Investigation The Temperature Dependence Of Schottky Barrier Diodes Characteristics Made Of Ga_2O_3

Thesis's outline:

The impact of growth temperature SBD performance using C-V-T and Zeta-theta method

The correlation between experimental and simulated capacitance-voltage at high-temperature conditions.

Ho Chi Minh University of Education – Bachelor – 2015 to 2019

- Department of Physic
- Major: Physics Teacher Education
- GPA: 3.25/4.0
- Graduation thesis: Time evolution of nuclear wave-packet in vibration molecules

Thesis's outline:

Solving the time-dependent of free Gauss, harmonic curve, and curve of oxygen with real potential

Wavepacket of oxygen and Frank Codon factor

Work experience

- Vo Truong Toan High School – Vietnam - Physics teacher - 2018
- Trang Education Video - Co-creator and Content Writer – 2019 to 2021
- Gia Dinh High School - Vietnam - Physics teacher - 2020
- Giang Hang education center - Vietnam - Physics teacher – 2020 to 2021
- Center of Soroban Linear thinking – Vietnam - Mathematics Teacher - 2019 to 2021
- High-tech Institute of Nguyen Tat Thanh University – Vietnam - Researcher – 2021 to 2022
- Korea Aerospace University – Republic of Korea - Researcher- 2022 to now

Academic Awards

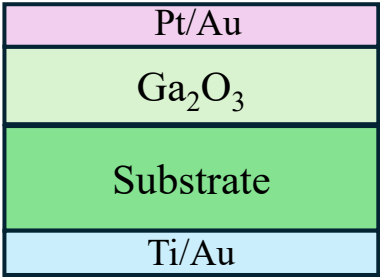
2024_ The 11th Asia-Pacific Workshop On Widebandgap Semiconductors_ Outstanding young researcher awards ([APWS 2024](#))

Title: Charge Neutral Point Shift Of $\alpha\text{-Ga}_2\text{O}_3$ Thin Film Upon Soft X-ray Irradiation

2025_ 11th Annual Conference of Vietnamese Young Scientists_ Best oral presentation ([Vietnam Young Scientists](#))

Title: Gallium Oxide: An Emerging Semiconductor for Next-Generation Power and Diode Applications

Journal Publications



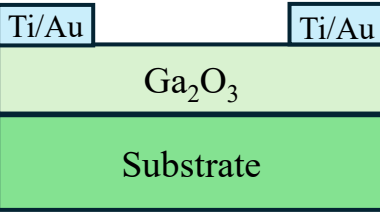
*Electrical characteristic

- Capacitance – voltage
- High temperature behavior
- Ultra-short pulse
- DC vs PC
- TCAD simulation

*Photo detector:

- X-ray response
- Application in low-Z detection

1. **2024/4/1_** Materials Science in Semiconductor Processing (IF 4.6)_ **1st author** _Temperature-dependent capacitance-voltage characteristics of β -Ga₂O₃ Schottky barrier diodes with (001) epitaxial grown layer using MOCVD
2. **2025/3/1_** Physica B: Condensed Matter (IF 2.8)_ **1st author** _ Correlation between experimental and modeled capacitance-voltage characteristics of Ga₂O₃ Schottky barrier diode in temperature range of 300–673 K
3. **2025/4/17_** ACS Applied Electronic Materials (IF 4.7)_ **1st author** _ Investigating Reverse Low-Bias Conduction Mechanisms in Ga₂O₃ Schottky Barrier Diodes Using Pulsed Current Methods
4. **2025/7/21_** Applied Physics Letters (IF 3.6)_ **1st author** _ High-sensitivity x-ray detection of stacked A4 sheets using Ga₂O₃ Schottky barrier diodes: Toward low-dose imaging and material discrimination
5. **2025/12/14_** Avanced Science (IF 14.1)_ **1st author** _ Direct Probing of Trap Dynamics in β -Ga₂O₃ Schottky Barrier Diodes Using Single-Voltage-Pulse Characterization
6. **2026_** Applied Physics Letters (IF 3.6)_ **1st author** _ Soft X-ray Shielding Characteristics of Natural Wood Films with Varying Densities Evaluated Using a Ga₂O₃ Schottky Diode Detector
7. **2025/12_** Applied Materials Today (IF 6.9)_ **co-author** _ Thermally stable X-ray detector using β -Ga₂O₃ schottky barrier diodes in extreme environments
8. **2026_** ACS photonics (IF 6.7)_ **co-author** _ Performance NiCo/ β -Ga₂O₃ Schottky Interfaces for Low-dose X-ray Detection and Imaging
9. **2026_** Progress in Quantum Electronics (IF 12.5)_ **1st author** _ Recent Advances in Ga₂O₃ Schottky Diodes for High-Temperature Power and Optoelectronic Devices



*Photo detector:

- X-ray response
- X-ray detector base Ga₂O₃ applications
- Wavelength-Dependent memory dynamics

1. **2024/7/26_** ACS Applied Electronic Materials (IF 4.7)_ **1st author** _Charge Neutral Point Shift of a 700 nm-Thick α -Ga₂O₃ Thin-Film Detector under Soft X-ray Irradiation
2. **2026_** APL_ Investigating the Impact of Source-to-Detector Distance via Neutral Point Modulation in Ga₂O₃ MSM X-ray Detectors (Submitted)

Conferences Publications

1. Time Dependence Of Charge Neutral Point Of Thin α - Ga₂O₃ Under 5 kV Soft X-ray. ACVYS , ISSN: 2093-4432 (2024)
2. Gallium Oxide: An Emerging Semiconductor for Next-Generation Power Electronics. ACVYS , ISSN: 2093-4432 (2025)